WORKSHEET

# Chapter 12: Substitution

**1** Evaluate:

**a** $y = 1 + 2x$ when $x = 0$

**b** $y = 3 + x$ when $x = 2$

**c** $y = 5x$ when $x = -1$

**d** $y = 3 + 2x$ when $x = 1$

**e** $y = 7 - x$ when $x = 5$

**f** $y = 4 + 3x$ when $x = 2$

**g** $y = 8 - 2x$ when $x = -1$

**h** $y = 1 - 5x$ when $x = 3$

**i** $y = 2 + 2x$ when $x = -4$

**j** $y = 3 - x$ when $x = -5$

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**k** $y = -4x$ when $y = -8$

**l** $y = 6 + 3x$ when $y = 9$

**m** $y = 4 + x$ when $y = 7$

**n** $y = 8 + 2x$ when $y = 4$

**o** $y = 3 - x$ when $y = 8$

**2** Substitute one equation into the other and simplify without solving:

**a** $y = 1$ and $y = 2 + 3x$

**b** $y = 2$ and $y = 1 + 3x$

**c** $y = 4$ and $y + x = 2$

**d** $x = 3$ and $y = 3 - 2x$

**e** $x = -1$ and $x + 2y = 3$

**f** $y = -3$ and $x + 2y = 6$

**g** $y = -1$ and $x - y = -2$

**h** $y = x + 1$ and $y = 3 + 2x$

**i** $y = 1 - x$ and $y = 6 + 2x$

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**j** $x = y + 1$ and $x + y = 7$ **k** $x = 1 - y$ and $y = 4 + 2x$ **l** $y = 4 - 3x$ and $y = 8 + x$

**m** $y = 7x$ and $y = 3 - x$ **n** $x = 3y - 2$ and $x + y + 2 = 0$ **o** $4x + 2y - 3 = 0$ and $y = 4 - 3x$

**3** Solve these simultaneous equations by substituting one equation into the other and simplifying:

**a** $y = 2$ and $y = 1 + x$ **b** $y = 3$ and $y = 2x$ **c** $y = -1$ and $y = 2 + x$

**d** $y = x$ and $y = 1 - x$ **e** $y = x$ and $y = 2x$ **f** $y = x$ and $y = 1 - 2x$

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**g** $y = x$ and $y = 3 - x$

**h** $y = -x$ and $y = 4 + x$

**i** $y = -x$ and $y = -1 + 2x$

**j** $y = -x$ and $y = 3x$

**k** $y = 1 + x$ and $y = 1 - x$

**l** $y = 1 + x$ and $y = 2x$

**m** $y = 1 - x$ and $y = 2x$

**n** $x + y = 8$ and $y = x$

**o** $x - y = 3$ and $y = 2$

**p** $2x + y - 1 = 0$ and $y = 1 - x$

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## Answers

**1** **a** $y = 1$ **b** $y = 5$ **c** $y = -5$
**d** $y = 5$ **e** $y = 2$ **f** $y = 10$
**g** $y = 10$ **h** $y = -14$ **i** $y = -6$
**j** $y = 8$ **k** $y = 32$ **l** $x = 1$
**m** $x = 3$ **n** $x = -2$ **o** $x = -5$

**2** **a** $1 = 2 + 3x$ **b** $2 = 1 + 3x$ **c** $4 + x = 2$
**d** $y = -3$ **e** $-1 + 2y = 3$ **f** $x - 6 = 6$
**g** $x + 1 = -2$ **h** $x + 1 = 3 + 2x$ **i** $1 - x = 6 + 2x$
**j** $2y + 1 = 7$ **k** $y = 6 - 2y$ **l** $4 - 3x = 8 + x$
**m** $7x = 3 - x$ **n** $4y = 0$ **o** $-2x + 5 = 0$

**3** **a** $(1, 2)$ **b** $(1.5, 3)$ **c** $(-3, -1)$ **d** $\left(\frac{1}{2}, \frac{1}{2}\right)$
**e** $(0, 0)$ **f** $\left(\frac{1}{3}, \frac{1}{3}\right)$ **g** $\left(1\frac{1}{2}, 1\frac{1}{2}\right)$ **h** $(-2, 2)$
**i** $\left(\frac{1}{3}, -\frac{1}{3}\right)$ **j** $(0, 0)$ **k** $(0, 1)$ **l** $(1, 2)$
**m** $\left(\frac{1}{3}, \frac{2}{3}\right)$ **n** $(4, 4)$ **o** $(5, 2)$ **p** $(0, 1)$

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